

Chyi Ricketts

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EDUCATION

Imperial College London

London, United Kingdom

September 2026

Master of Research in Cancer Informatics

- **Key Project:** Novel Biomarker Discovery for Mesenchymal Glioblastomas via Multi-Omics Analysis
- **Thesis:** Target-Specific Binding Affinity Prediction Scoring Functions for Drug Target Identification in Lead Optimization

Duke University & Duke Kunshan University (DKU) Dual Degree

May 2025

Durham, North Carolina, USA & Kunshan, Jiangsu, China

Bachelor of Science in Molecular Bioscience with a Track in Genetics and Genomics

- **Honors:** *Graduation with Distinction* | GPA: 3.7/4.0
- **Relevant Coursework:** Comp. Genomics, Comp. Biology & Bioinformatics, Adv. Methods in Functional Genomics
- **Thesis:** Integrating Segmentation Software for Automated Cellular Image Analysis in a Web-Based Application

LABORATORY SKILLS

Computational and Bioinformatics Tools

Python (data analysis, bioinformatics, ML)
R / RStudio (statistical analysis, visualization)
Other Languages: SQL, HTML/CSS, Java, JS
Workflow & reproducibility: Git/GitHub, Docker, Nextflow
Sequence analysis: BLAST, bwa, kallisto, MMseqs2, TM-align
Machine learning: scikit-learn, PyTorch, TensorFlow
Image analysis: ImageJ/Fiji, Cellpose, Segment Anything
Systems: Linux (command line), HPC (PBS/qsub), AWS

Wet Lab Techniques

Mammalian and bacterial cell culture
DNA/RNA extraction and purification
Prokaryotic siRNA production and purification
PCR and RT-qPCR
Western blotting
CRISPR/Cas9 gene editing
Confocal & fluorescence microscopy
Zebrafish embryo microinjection

RELEVANT RESEARCH EXPERIENCE

Imperial College London ~ *Ballester Lab*

Nov. 2025 – Sep. 2026

London, United Kingdom

Project: Multi-Omics Identification of a Novel Prognostic Biomarker in Glioblastoma Subtypes

- Applied high-throughput multi-omics (RNA-seq, DNA methylation, CNV, RPPA) to TCGA glioblastoma cohorts (n=600+), performing Cox regression and Kaplan-Meier survival analyses to nominate ARSI as a candidate prognostic biomarker
- Identified subtype-specific survival associations in the Mesenchymal GBM subtype and characterized epigenetic upregulation of ARSI via CpG hypomethylation analysis integrating molecular and clinical covariates to identify subtype-specific prognostic signatures, providing experience with the challenges of multi-modal data integration in disease cohorts

Conducting Masters thesis research on ML scoring functions for cancer drug discovery across five protein targets

- Curated and preprocessed large-scale protein–ligand datasets (~100,000 complexes) with RDKit and ECFP6 Tanimoto FPs
- Designed and maintained reproducible, large-scale computational benchmarking pipelines on HPC infrastructure, with emphasis on modular workflow design and cross-dataset evaluation for 100+ protein-ligand models across featurization schemes (AEV-PLIG, ECIF, PLEC) and learning algorithms (GATv2, XGBoost, Random Forest, Support Vector Machine)

Duke Kunshan University ~ *Huang Lab*

May 2023 – May 2025

Kunshan, Jiangsu, China

Project: Computational benchmarking of image segmentation tools for biological microscopy

- Quantitatively benchmarked generic and fine-tuned Cellpose, Ilastik, and SAM on self-generated RNAi-based HeLa cell imaging data using custom computational metrics, following qualitative screening of candidate segmentation tools
- Developed an interactive visualization tool for post-segmentation image analysis (<https://sw-cellvis.onrender.com/>)

Grant writing: Authored and submitted a successful grant proposal to the Kunshan Supercomputing Center, resulting in the award of 9 high-performance computing nodes to support image analysis and segmentation software benchmarking

Duke University ~ *Di Talia Lab & Bagnat Lab*

April 2022 – July 2024

Durham, North Carolina, USA

Project: DST (Duke Science and Technology) SPARK Project: “*Synchronized Clocks in Zebrafish Patterning*”

- Trained and benchmarked 10 Cellpose models analyze oscillating ERK expression in embryonic zebrafish notochords
- Applied Knock-in Tagging methods to examine protein participation in nutrient transport within the interstitial lumen

ADDITIONAL SKILLS

- **Languages:** Native English Speaker, Fluent in Mandarin and Advanced literacy in Traditional and Simplified Chinese